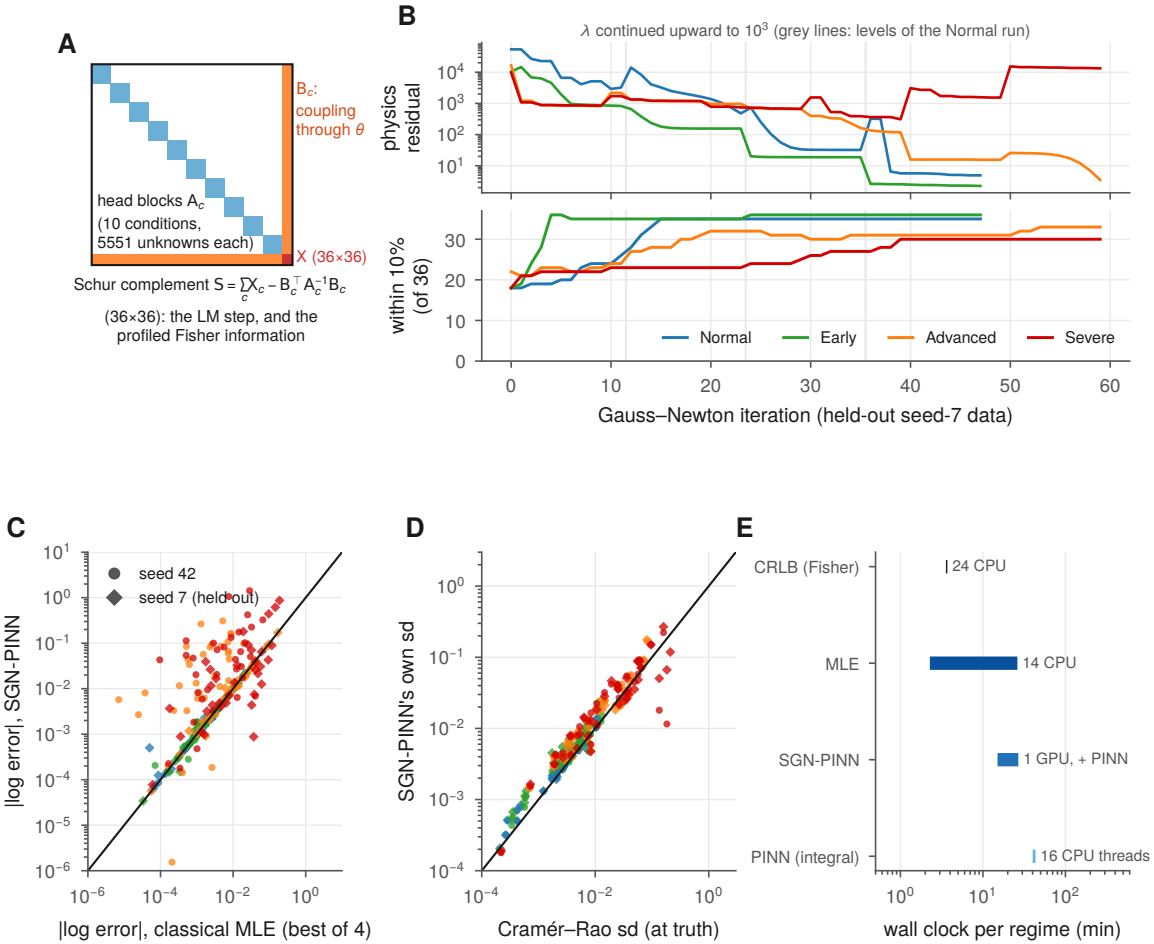

A
B_c: coupling through θ
head blocks A_c
(10 conditions, 5551 unknowns each)
X (36×36)
Schur complement S = ∑_c X_c − B_c^⊤ A_c^{-1} B_c
(36×36): the LM step, and the profiled Fisher information
B
λ continued upward to 10^3 (grey lines: levels of the Normal run)
physics residual
within 10% (of 36)
Normal
Early
Advanced
Severe
Gauss–Newton iteration (held-out seed-7 data)
C
seed 42
seed 7 (held-out)
|log error|, SGN-PINN
|log error|, classical MLE (best of 4)
D
SGN-PINN's own sd
Cramér–Rao sd (at truth)
E
CRLB (Fisher)
24 CPU
MLE
14 CPU
SGN-PINN
1 GPU, + PINN
PINN (integral)
16 CPU threads
wall clock per regime (min)